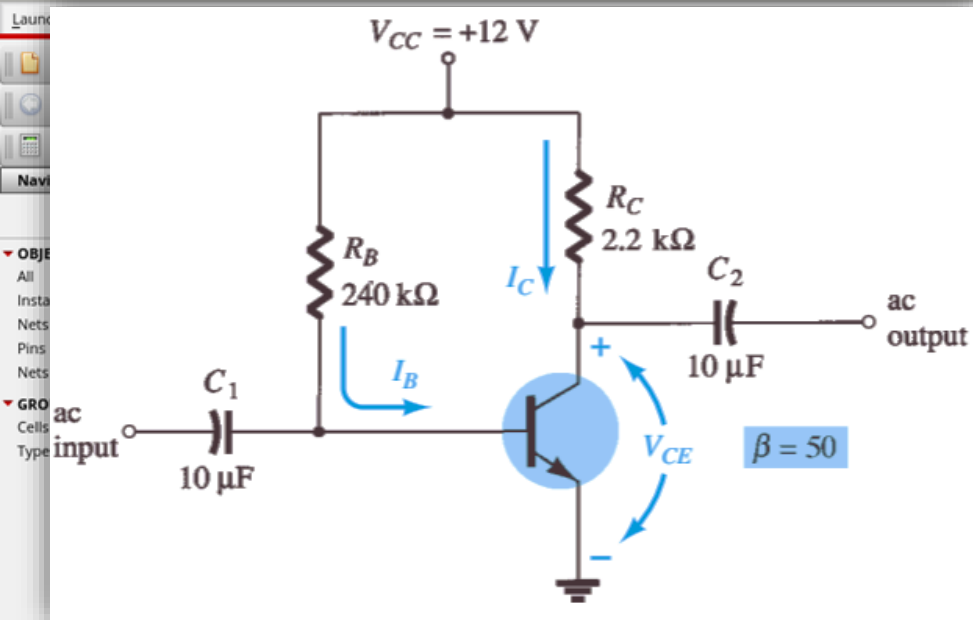


BJT D.C ANALYSIS

Er. Sailesh Mishra

B.Tech (ECE | Silicon University)

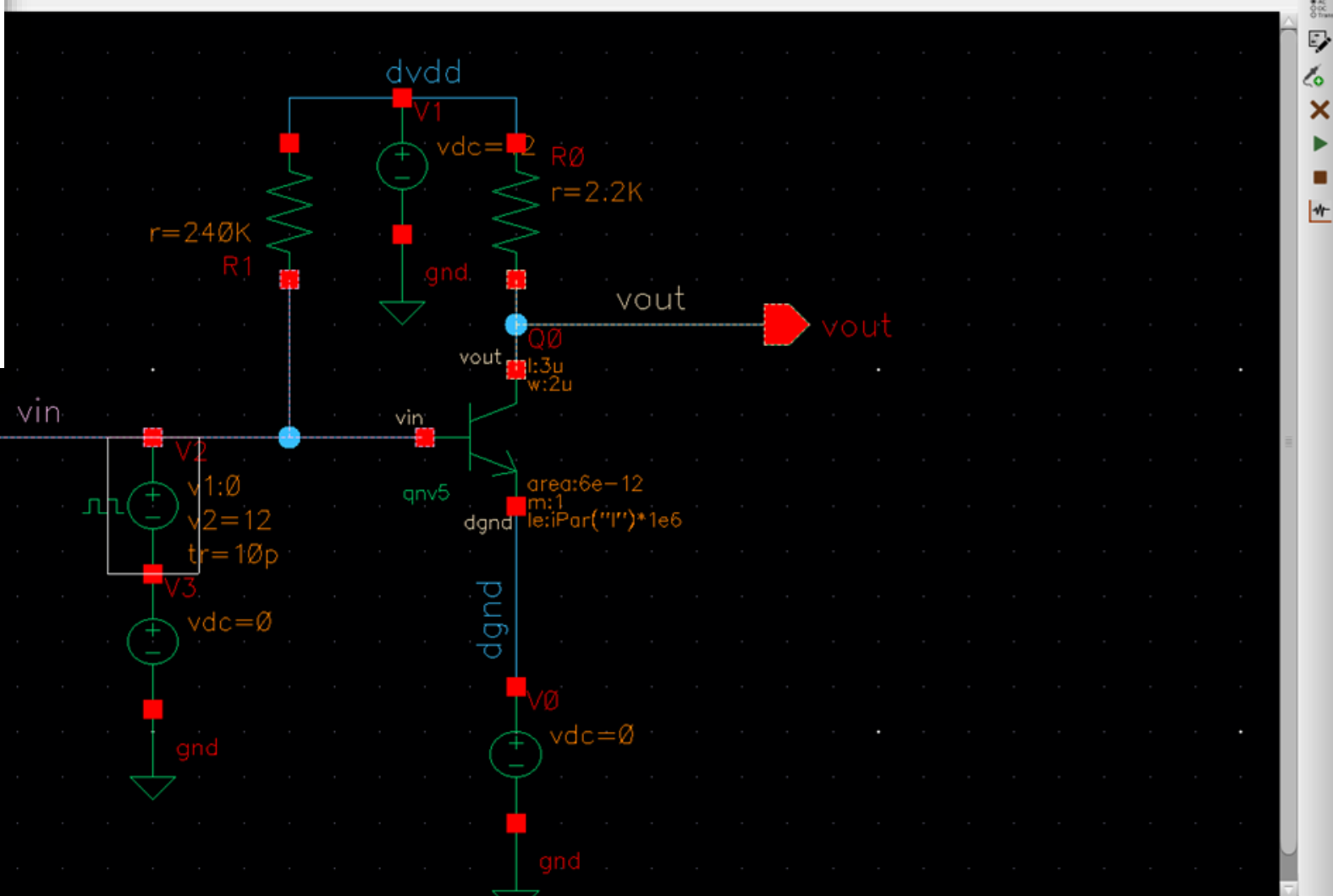
Specialization in Microelectronics & VLSI



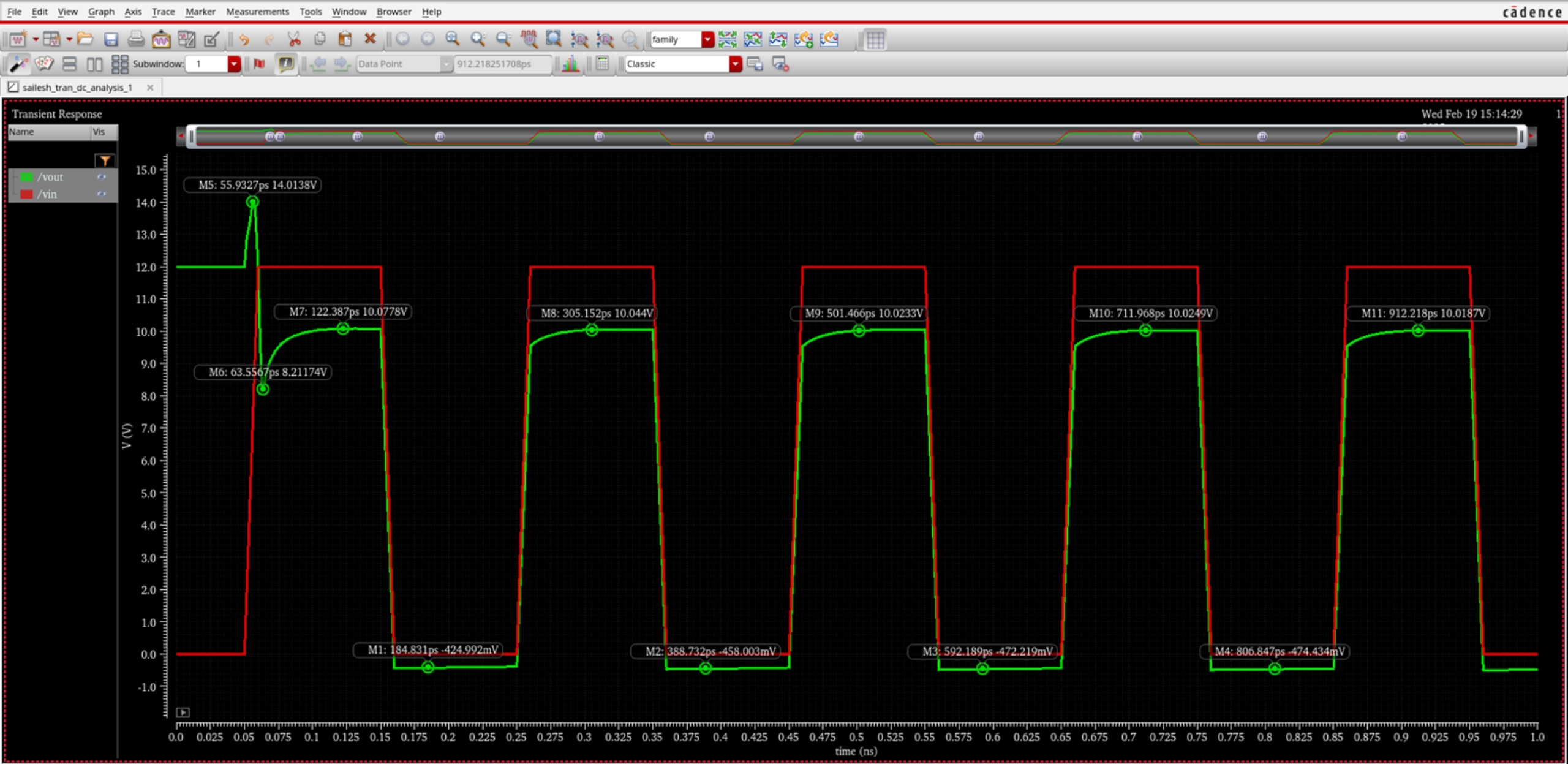
FIXED BIAS CIRCUIT WITHOUT COUPLING CAPACITORS

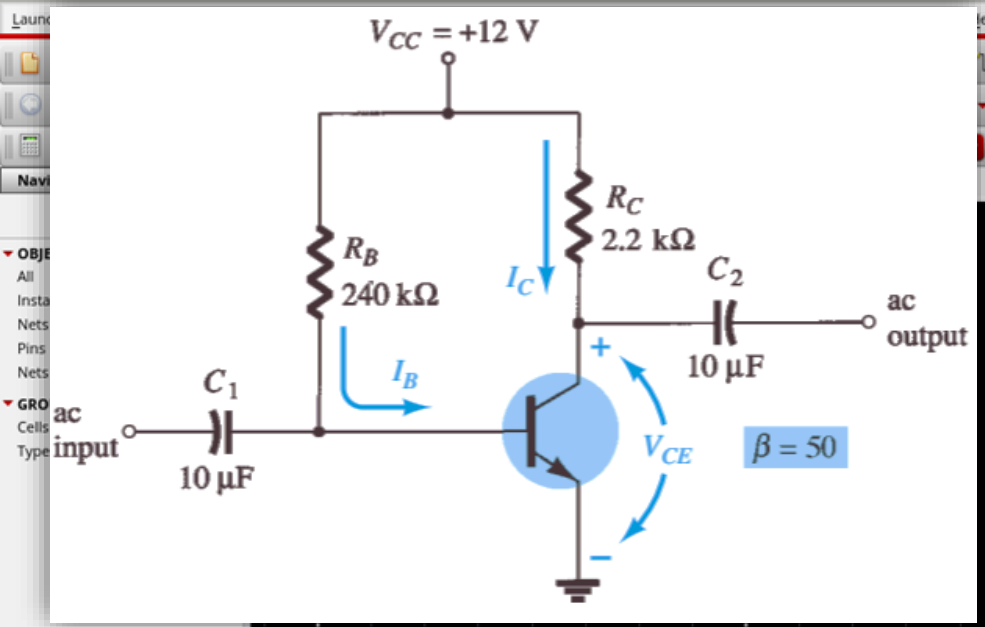
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XF magnitude	
PAC magnitude	
PAC phase	
Voltage 1	0 V
Voltage 2	12 V
Period	200p s



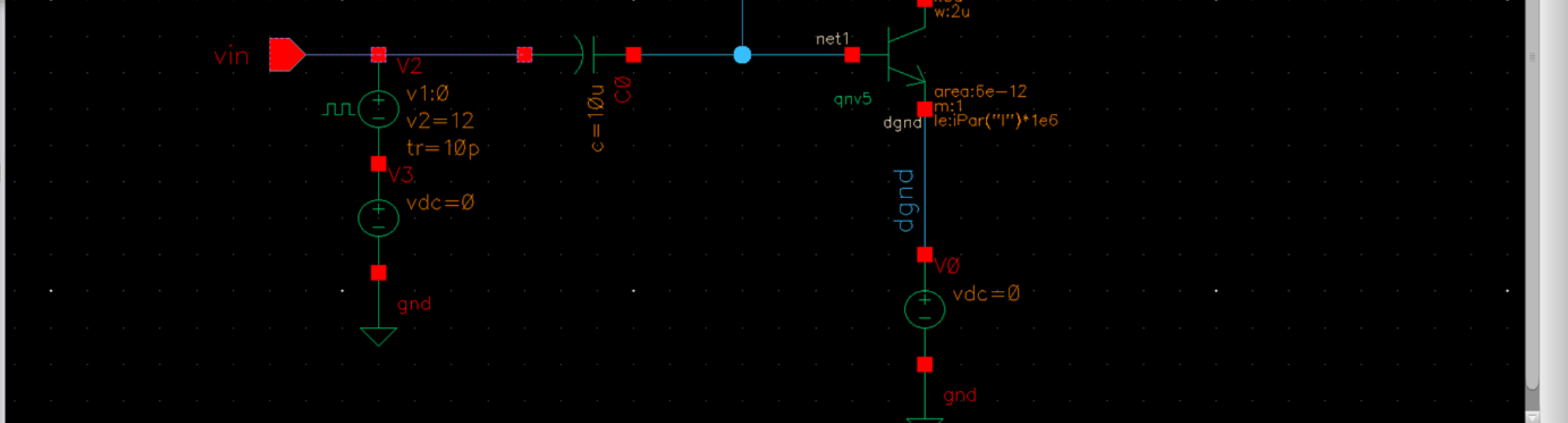
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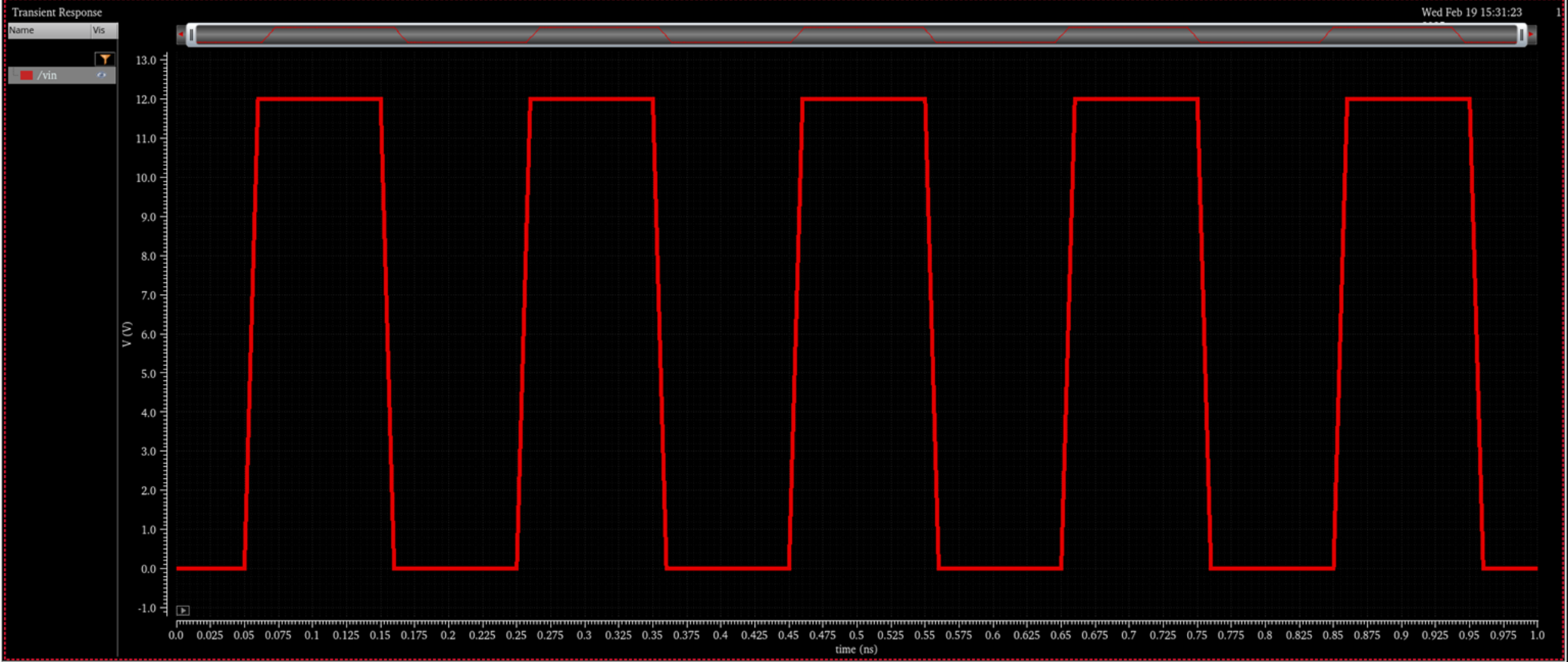
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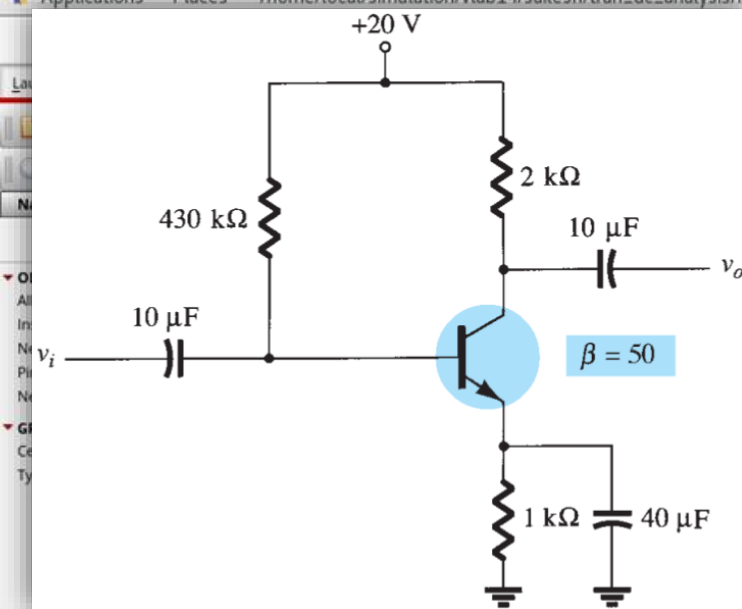
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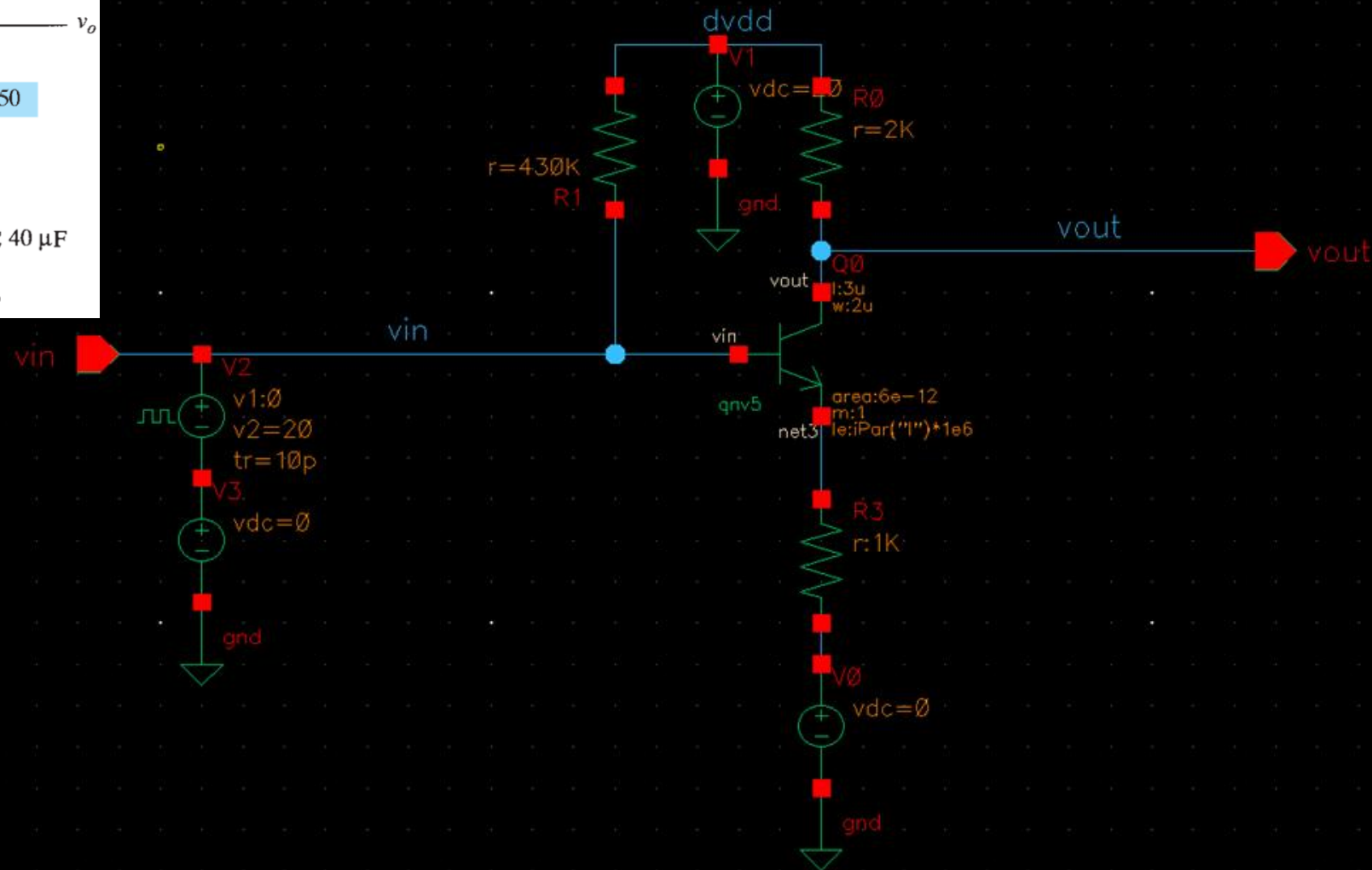
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Virtuoso® Schematic Editor L Editing: sailesh tran_dc_analysis schematic

EMITTER BIAS CIRCUIT WITHOUT COUPLING CAPACITORS



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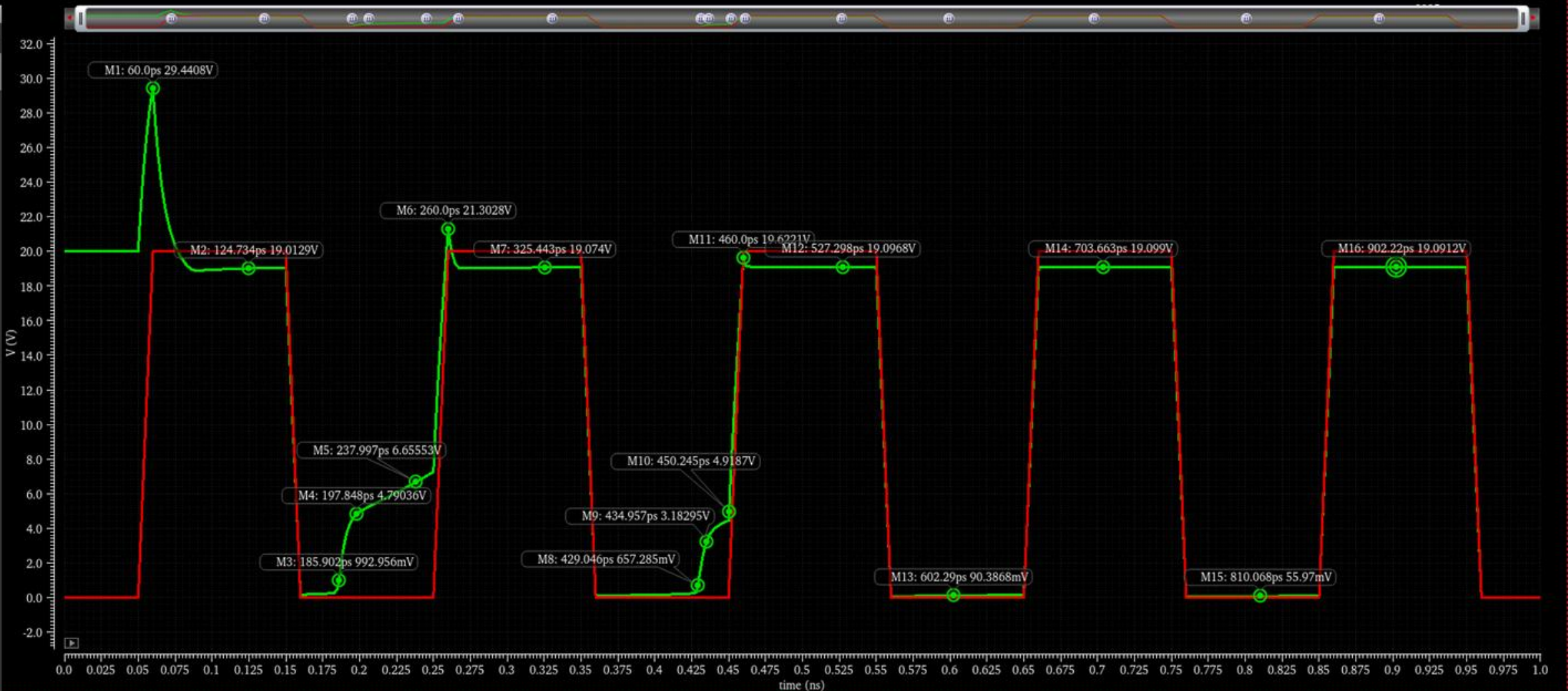


sailesh_tran_dc_analysis_1

Transient Response

Name Vis

☒ /vout
☒ /vin

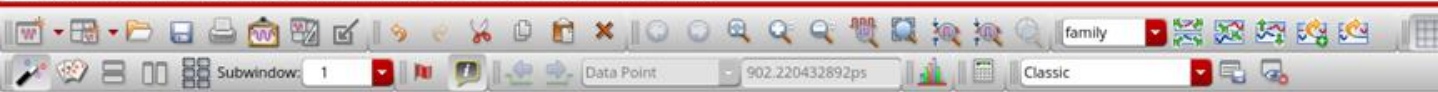


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M:

R:

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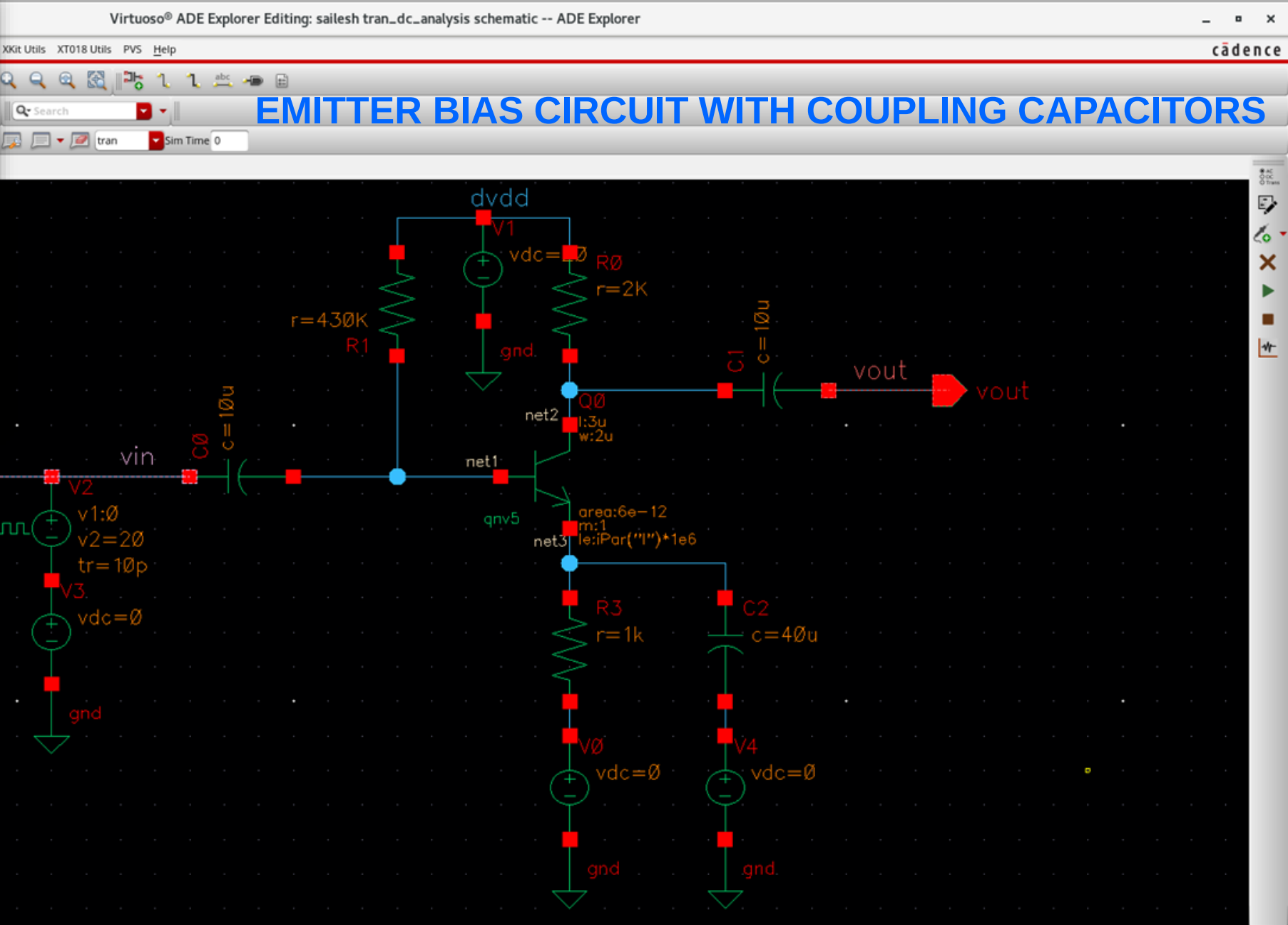
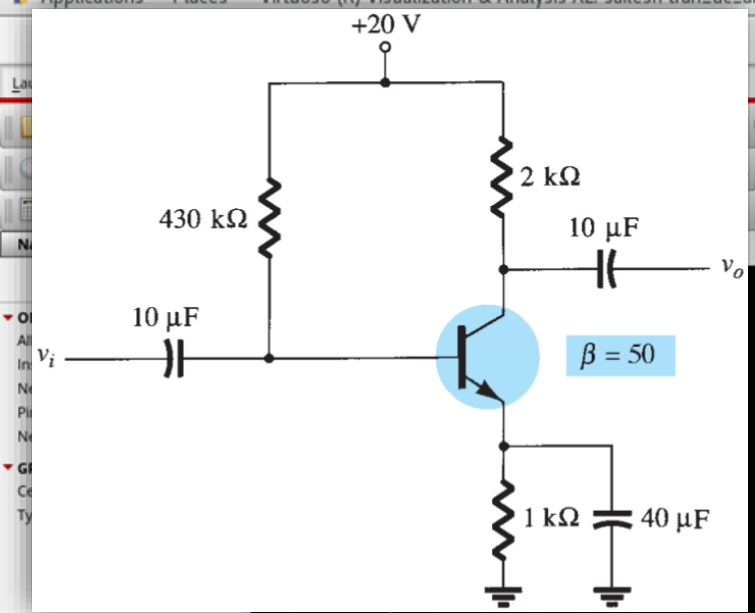


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sailesh_tran_dc_analysis_1



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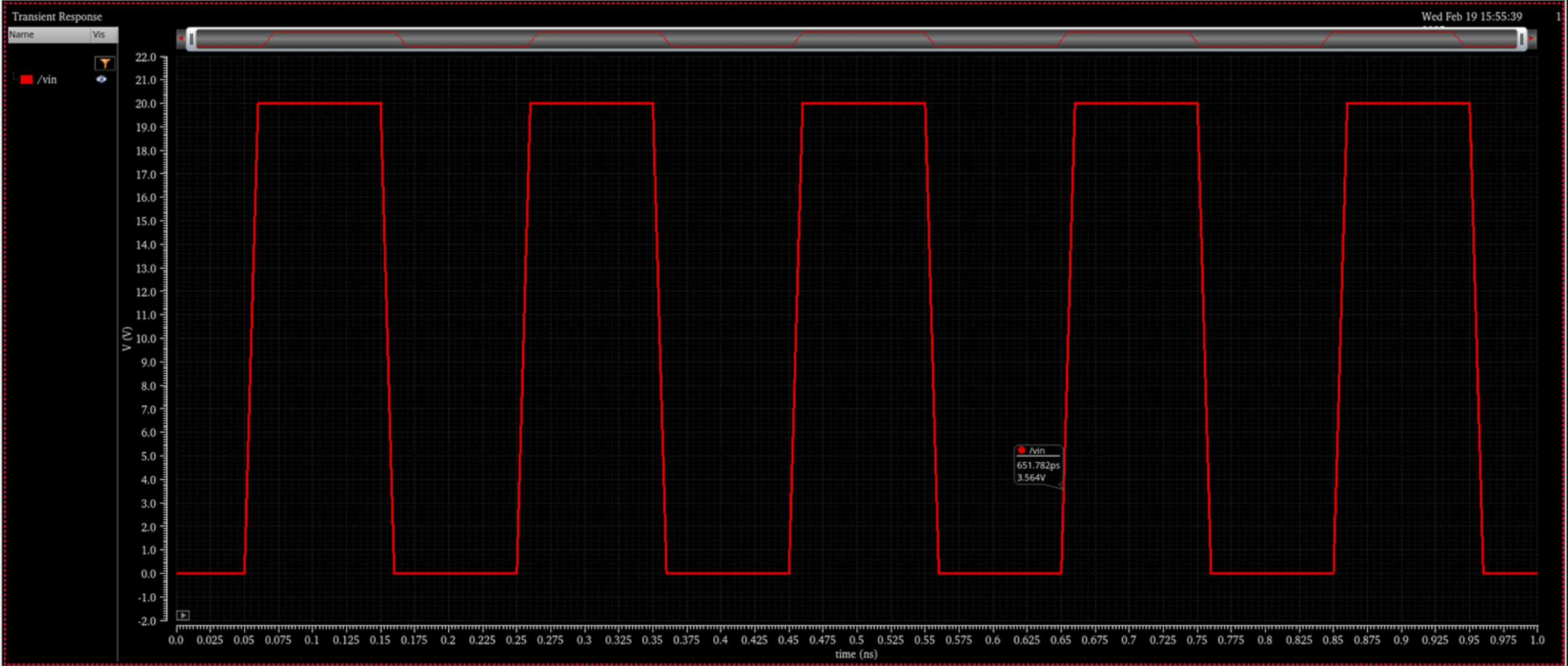
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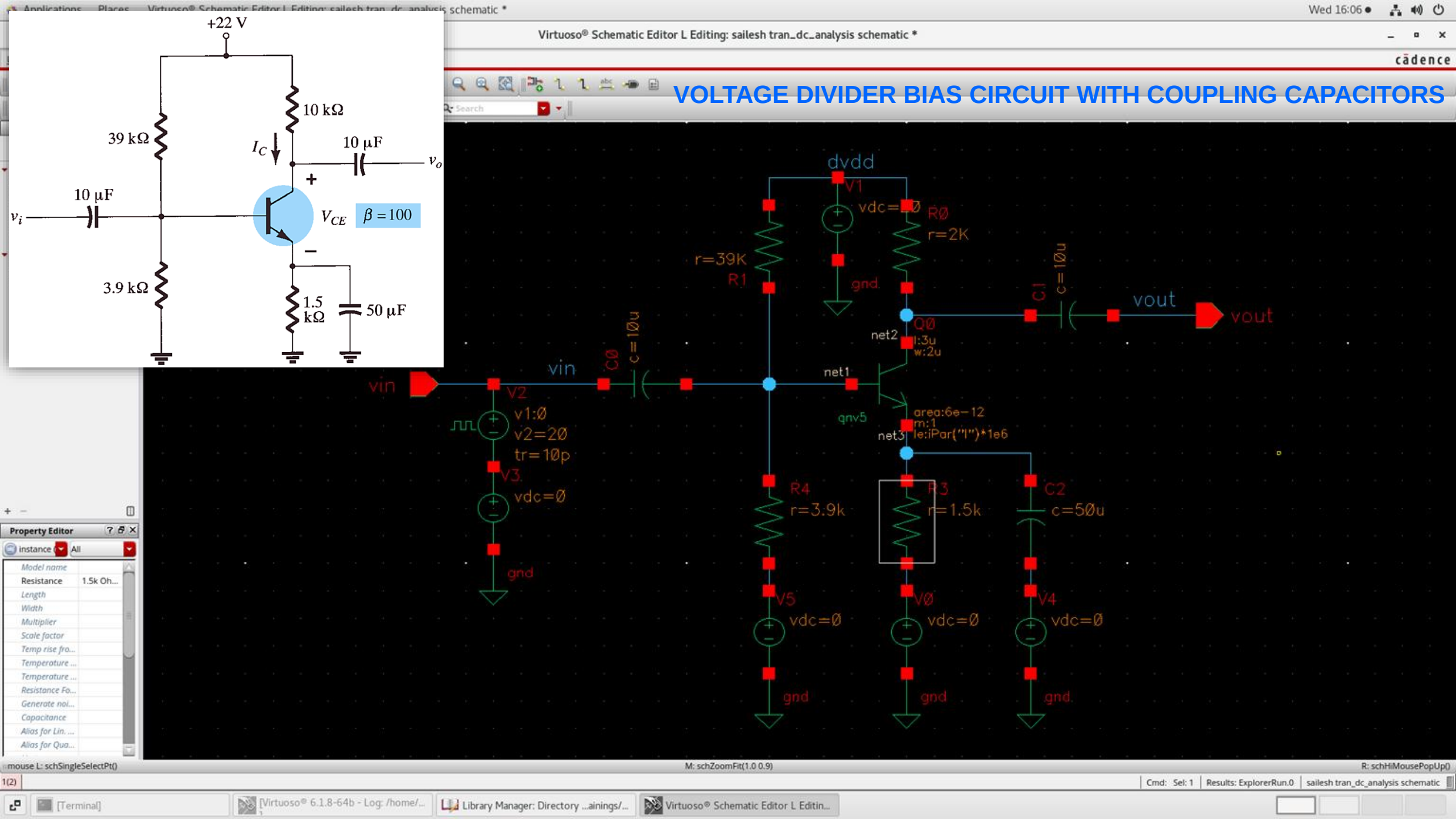
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Units	inch

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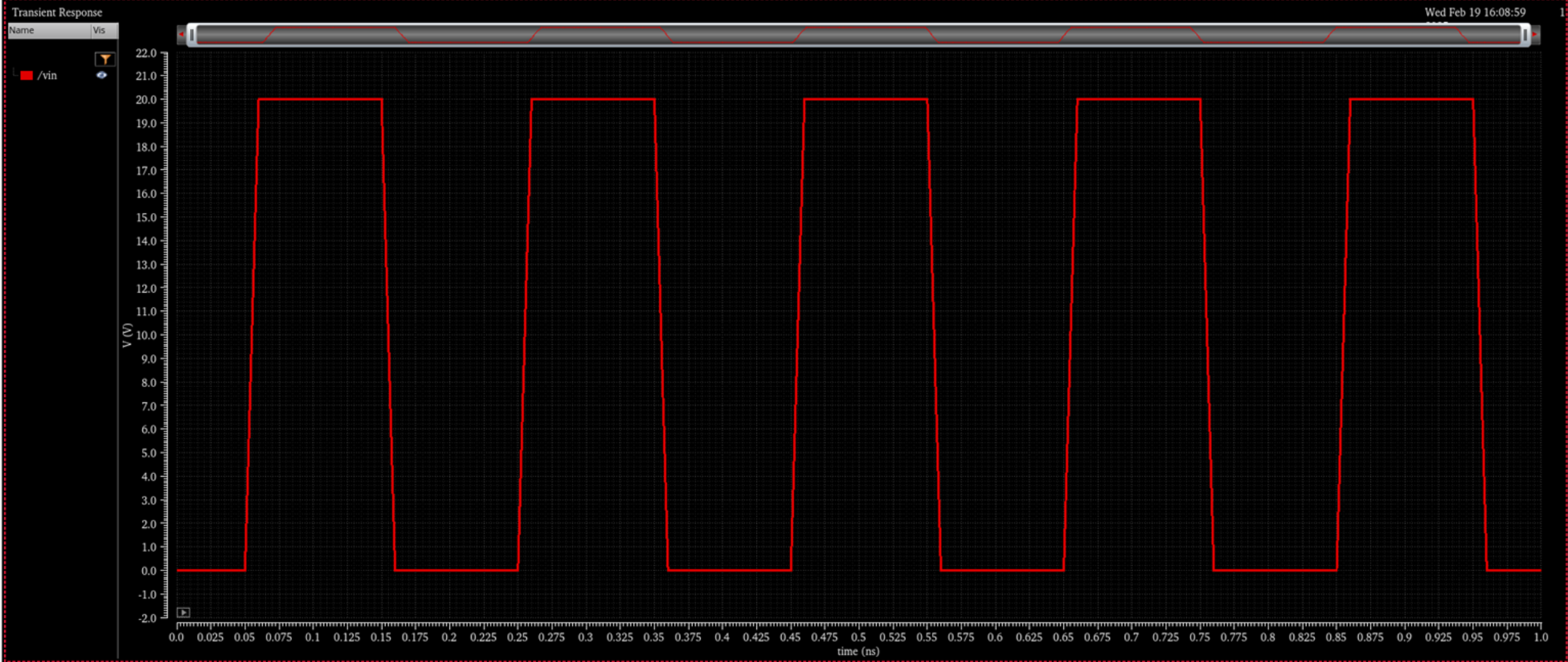
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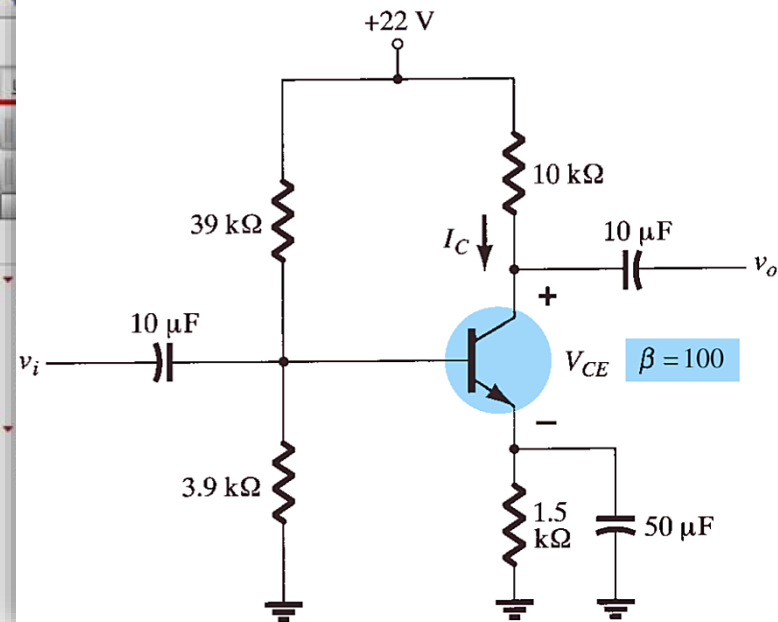




sailesh_tran_dc_analysis_1 x



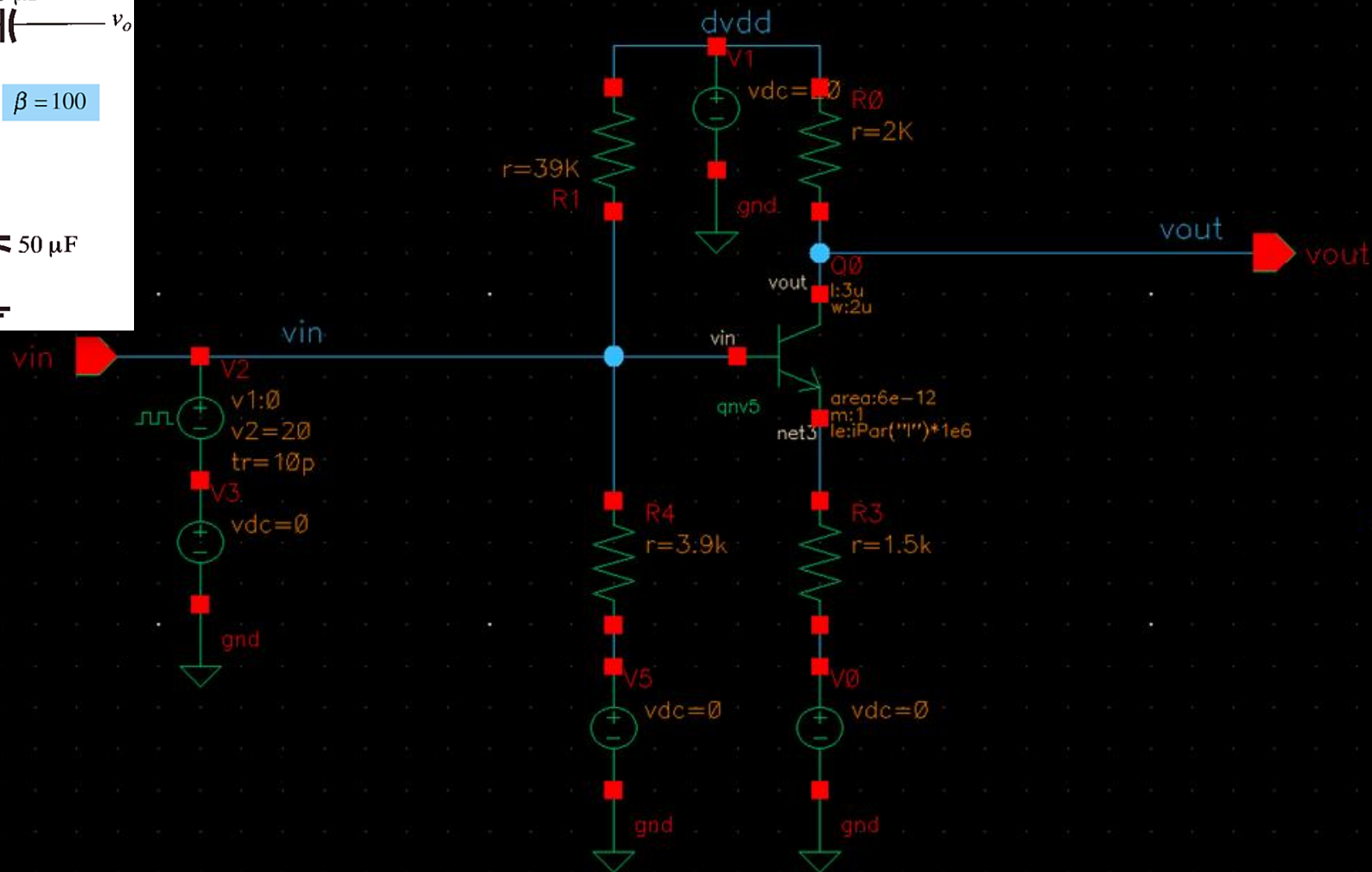
VOLTAGE DIVIDER BIAS CIRCUIT WITHOUT COUPLING CAPACITORS



Property Editor

Schematic Attributes

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Units	inch





sailesh_tran_dc_analysis_1



9(37) Trace: Avout; History: ExplorerRun.0; Test: sailesh_tran_dc_analysis_1; Design Point: 1; Corner: nom

Virtuoso (R) Visualization & Analysis XL: sailesh tran_dc_analysis maistro

File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help

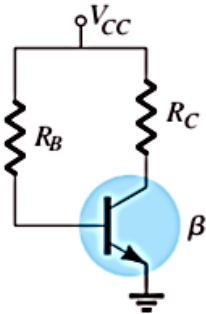
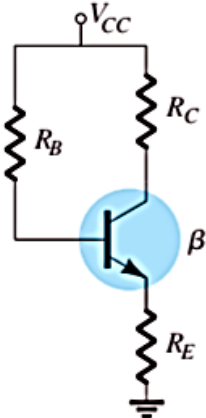
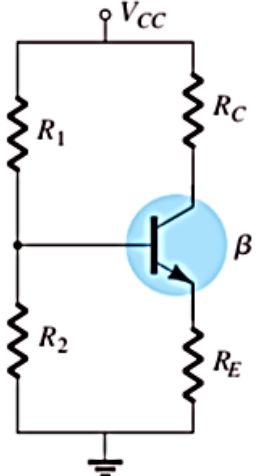
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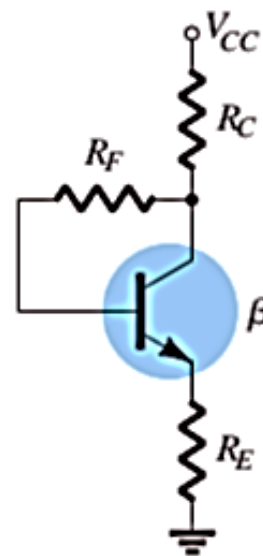
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TABLE 4.1
BJT Bias Configurations

Type	Configuration	Pertinent Equations
Fixed-bias		$I_B = \frac{V_{CC} - V_{BE}}{R_B}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $V_{CE} = V_{CC} - I_C R_C$
Emitter-bias		$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $R_i = (\beta + 1)R_E$ $V_{CE} = V_{CC} - I_C(R_C + R_E)$
Voltage-divider bias		EXACT: $R_{Th} = R_1 \parallel R_2, E_{Th} = \frac{R_2 V_{CC}}{R_1 + R_2}$ $I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $V_{CE} = V_{CC} - I_C(R_C + R_E)$ APPROXIMATE: $\beta R_E \geq 10R_2$ $V_B = \frac{R_2 V_{CC}}{R_1 + R_2}, V_E = V_B - V_{BE}$ $I_E = \frac{V_E}{R_E}, I_B = \frac{I_E}{\beta + 1}$ $V_{CE} = V_{CC} - I_C(R_C + R_E)$

Collector-feedback

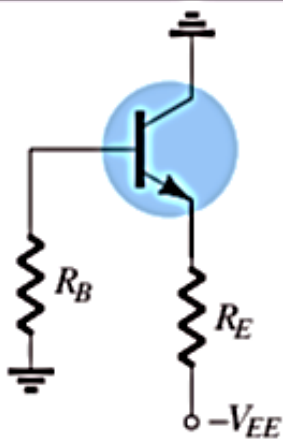


$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta(R_C + R_E)}$$

$$I_C = \beta I_B, I_E = (\beta + 1)I_B$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

Emitter-follower

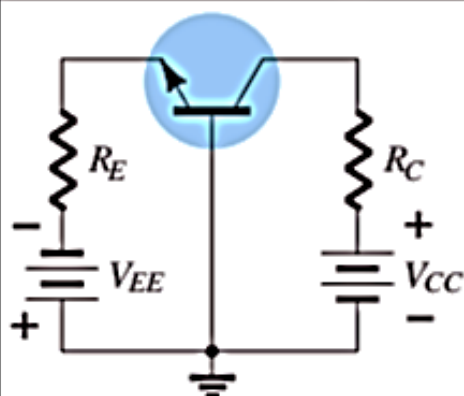


$$I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E}$$

$$I_C = \beta I_B, I_E = (\beta + 1)I_B$$

$$V_{CE} = V_{EE} - I_E R_E$$

Common-base



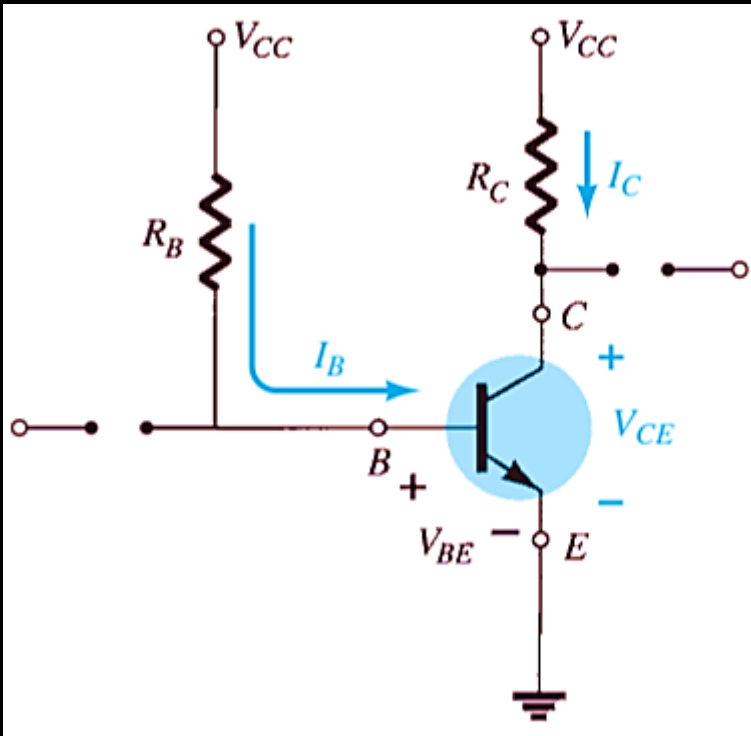
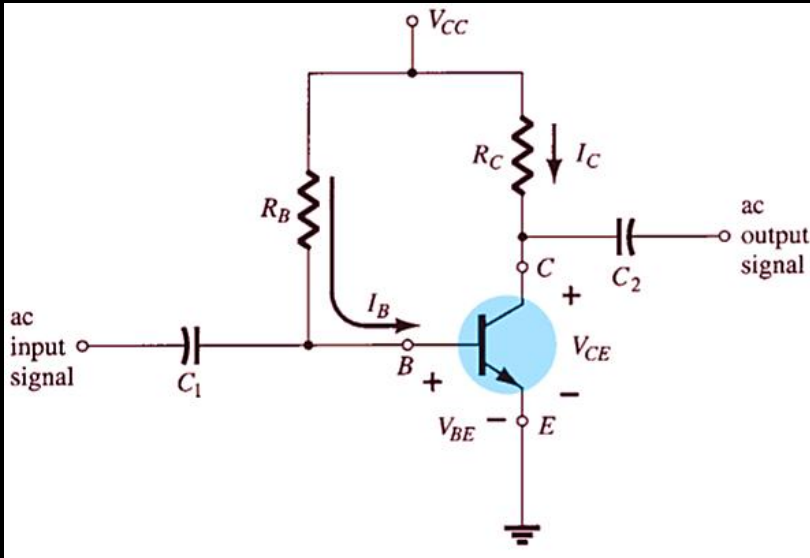
$$I_E = \frac{V_{EE} - V_{BE}}{R_E}$$

$$I_B = \frac{I_E}{\beta + 1}, I_C = \beta I_B$$

$$V_{CE} = V_{EE} + V_{CC} - I_E(R_C + R_E)$$

$$V_{CB} = V_{CC} - I_C R_C$$

1. FIXED BIAS conf. [DC BIASING]



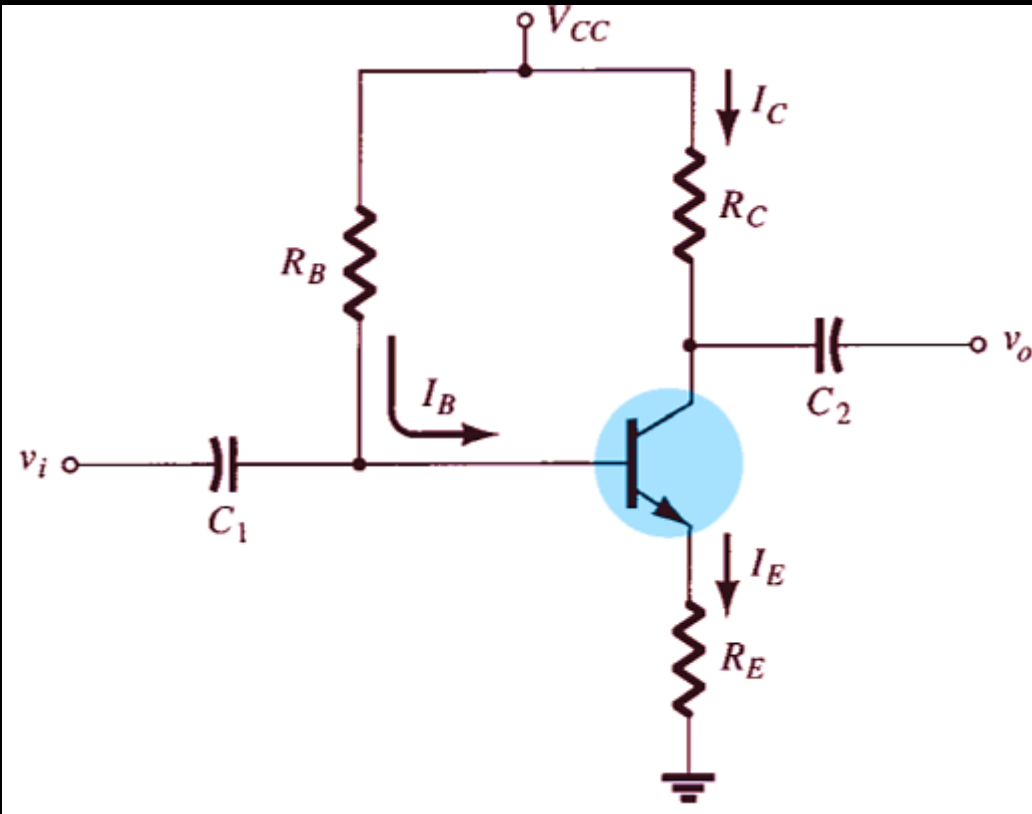
❑ Limitations :-

- **Thermal Runaway:** The most significant drawback of fixed bias is its susceptibility to thermal variations. BJT's base-emitter voltage (V_{be}) exhibits a strong dependence on temperature. As temperature increases, V_{be} decreases, leading to a rise in collector current (I_C). This positive feedback loop can cause thermal runaway, where I_C continues to climb, further increasing temperature and potentially damaging the transistor. Careful component selection and heat sinking become essential for mitigating this effect.
- **Power Inefficiency:** The biasing resistor (R_B) in a fixed bias configuration constantly draws current, resulting in wasted power dissipation. This becomes a major concern in battery-powered or low-power circuits where efficient operation is paramount.

❑ Advantages :-

- **Simplicity:** Fixed bias is the most straightforward biasing method, requiring only a single resistor for operation. This makes it ideal for basic amplifier circuits, rapid prototyping, or educational demonstrations.
- **Suitable for Low-Frequency Applications:** When temperature variations are minimal and power consumption isn't a critical factor, fixed bias can be a viable option for low-frequency circuits where stability is less of a concern.

2. EMITTER BIAS conf. [DC BIASING]



❑ Limitations : -

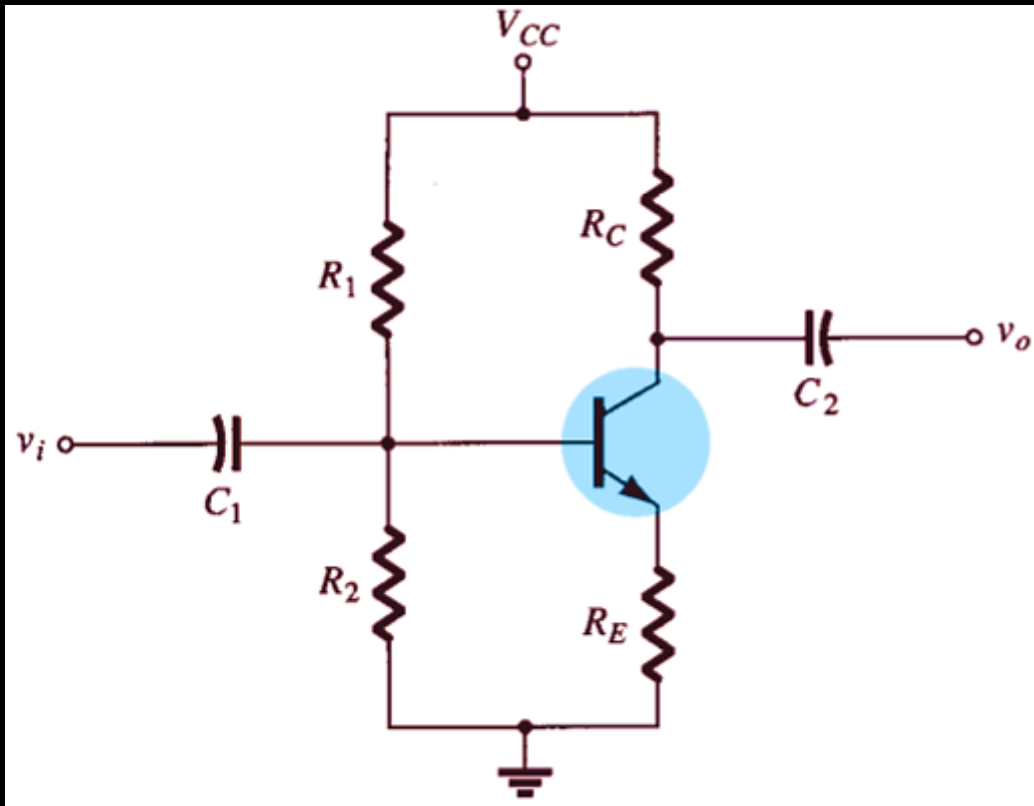
- **Moderate Temperature Dependence:** While offering improved stability over fixed bias, I_C in emitter bias still exhibits some temperature dependence due to changes in V_{be} . This is because the biasing resistor (R_E) introduces a voltage drop across itself, affecting the base-emitter voltage and consequently I_C . Thermal considerations remain important for ensuring consistent circuit performance.
- **Limited Control over I_C :** The value of R_E sets I_C indirectly. Transistor parameters like beta (current gain) can introduce variations in the actual collector current, making it challenging to achieve precise control over I_C in critical applications.

❑ Advantages : -

- **Simpler Implementation:** Emitter bias requires fewer components compared to voltage divider biasing, simplifying circuit design and potentially reducing board space and cost.
- **Enhanced Stability:** The self-biasing effect of R_E in emitter bias helps to stabilize I_C against temperature fluctuations to a greater extent compared to fixed bias. This leads to improved circuit performance in terms of gain and distortion characteristics.

3a. VOLTAGE DIVIDER BIAS conf. [exact analysis]

[DC BIASING]



Exact Method - > (using 3 resistors)

□ Limitations:

- **Increased Component Count:** The exact voltage divider method utilizes three resistors for biasing, potentially affecting board space and cost, especially in space-constrained applications.
- **Residual Temperature Dependence:** While significantly improved, I_c might still exhibit some residual temperature dependence due to the inherent characteristics of BJTs. This necessitates considerations for circuits requiring exceptionally high stability across a wide temperature range.

□ Advantages:

- **Superior Stability:** The exact voltage divider method provides the most stable operating point of all the discussed configurations. By incorporating an additional resistor, it effectively minimizes the influence of V_{be} variations on I_c , leading to consistent performance across temperature variations.
- **Precise Control over I_c :** This method allows for the most accurate setting and control of collector current, making it crucial for applications requiring precise amplification characteristics, such as linear amplifiers or operational amplifiers.

3b. VOLTAGE DIVIDER BIAS conf. [approximate analysis]

[DC BIASING]

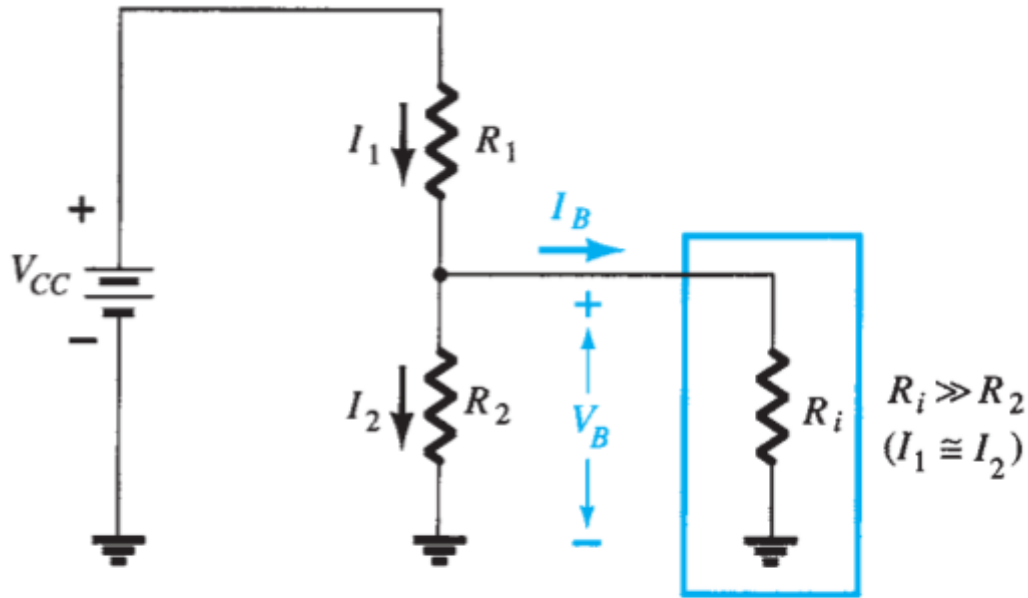


FIG. 4.36

Partial-bias circuit for calculating the approximate base voltage V_B .

Approximate Method - > (using 2 resistors)

□ Limitations : -

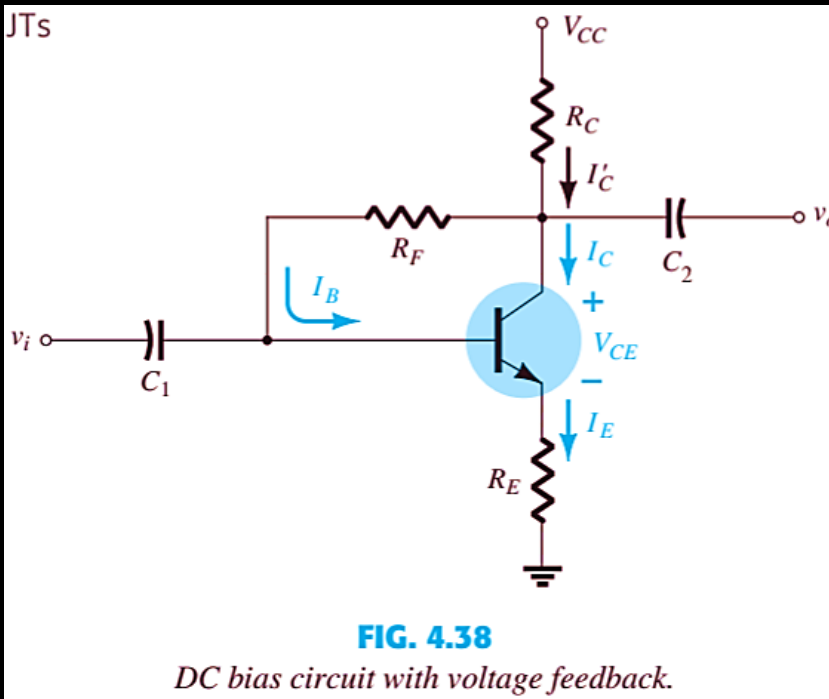
- **Temperature Dependence:** Similar to emitter bias, the approximate voltage divider method exhibits moderate temperature dependence in I_C due to the influence of V_{BE} on the voltage across R_E . This necessitates careful consideration in thermally sensitive applications.
- **Moderate Control over I_C :** The resistor ratio sets I_C approximately, but variations in transistor parameters can introduce slight deviations from the calculated value. Achieving highly precise control over I_C might be challenging with this method.

□ Advantages : -

- **Improved Stability:** Compared to fixed and emitter bias, the approximate voltage divider method offers a more stable operating point for the BJT. This leads to better circuit performance in terms of consistent gain and reduced distortion.
- **Simpler Design:** This method requires fewer components for basic biasing, minimizing board space and cost compared to the exact method.

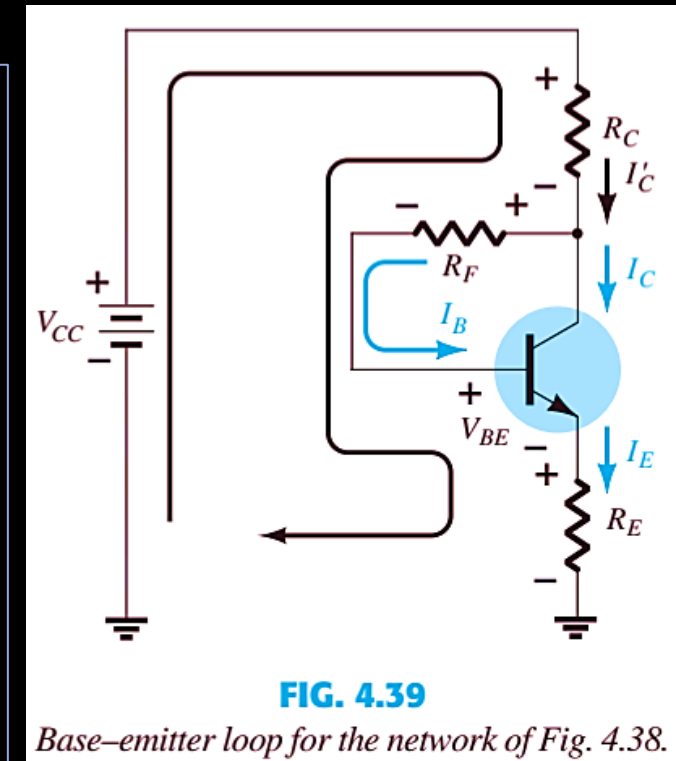
4. COLLECTOR FEEDBACK conf.

[DC BIASING]



Limitations :

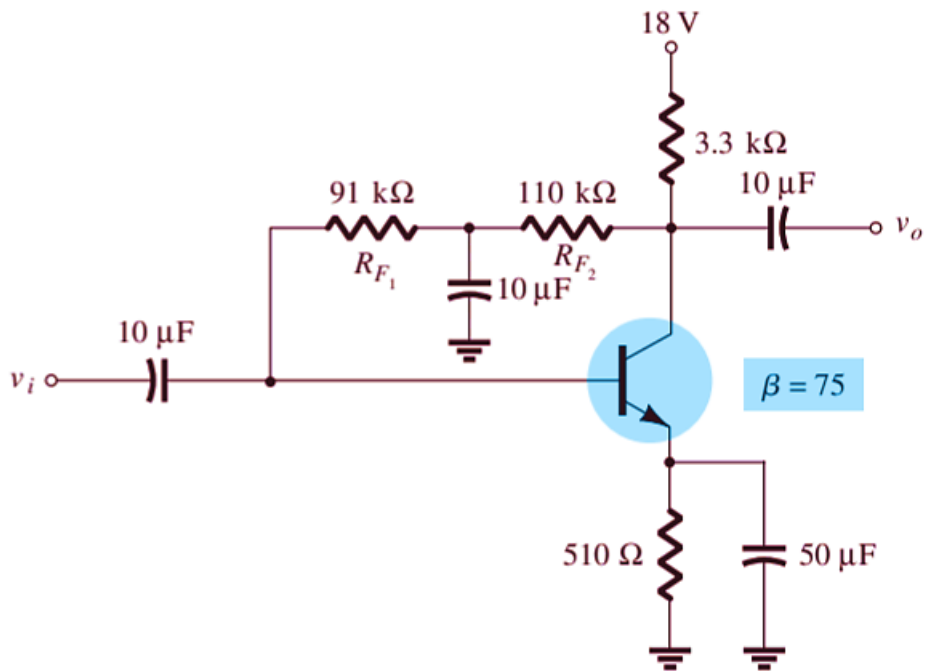
- **Reduced Gain (Continued)** : As mentioned earlier, collector feedback introduces negative feedback into the circuit. While this enhances stability, it also reduces the overall voltage gain of the amplifier. This trade-off between stability and gain needs to be carefully evaluated for the specific application. High-gain voltage amplifiers might not be suitable candidates for collector feedback biasing.
- **Increased Design Complexity** : - Collector feedback biasing requires additional components (a resistor and a capacitor) compared to simpler methods. Careful selection of component values and design analysis are essential to ensure proper biasing, frequency response, and stability. Miscalculations or improper component selection can lead to undesirable circuit behavior.



Advantages:

- **Superior Stability** : Collector feedback offers excellent stability for the BJT's operating point. The negative feedback mechanism helps to counteract variations in transistor parameters and temperature fluctuations, leading to consistent performance in terms of gain and distortion. This makes it ideal for applications requiring high precision and minimal drift, such as instrumentation amplifiers or reference voltage circuits.
- **Reduced Distortion** : The negative feedback also helps to minimize distortion in the amplified signal by suppressing the effects of non-linearity in the BJT's characteristics. This is crucial for applications where maintaining a clean, undistorted output is critical, such as audio amplifiers or pre-amplifiers.

Illustration (I) - >



Solution: In this case, the base resistance for the dc analysis is composed of two resistors with a capacitor connected from their junction to ground. For the dc mode, the capacitor assumes the open-circuit equivalence, and $R_B = R_{F1} + R_{F2}$.

Solving for I_B gives

$$\begin{aligned} I_B &= \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} \\ &= \frac{18 \text{ V} - 0.7 \text{ V}}{(91 \text{ k}\Omega + 110 \text{ k}\Omega) + (75)(3.3 \text{ k}\Omega + 0.51 \text{ k}\Omega)} \\ &= \frac{17.3 \text{ V}}{201 \text{ k}\Omega + 285.75 \text{ k}\Omega} = \frac{17.3 \text{ V}}{486.75 \text{ k}\Omega} \\ &= 35.5 \mu\text{A} \end{aligned}$$

$$I_C = \beta I_B$$

$$= (75)(35.5 \mu\text{A})$$

$$= 2.66 \text{ mA}$$

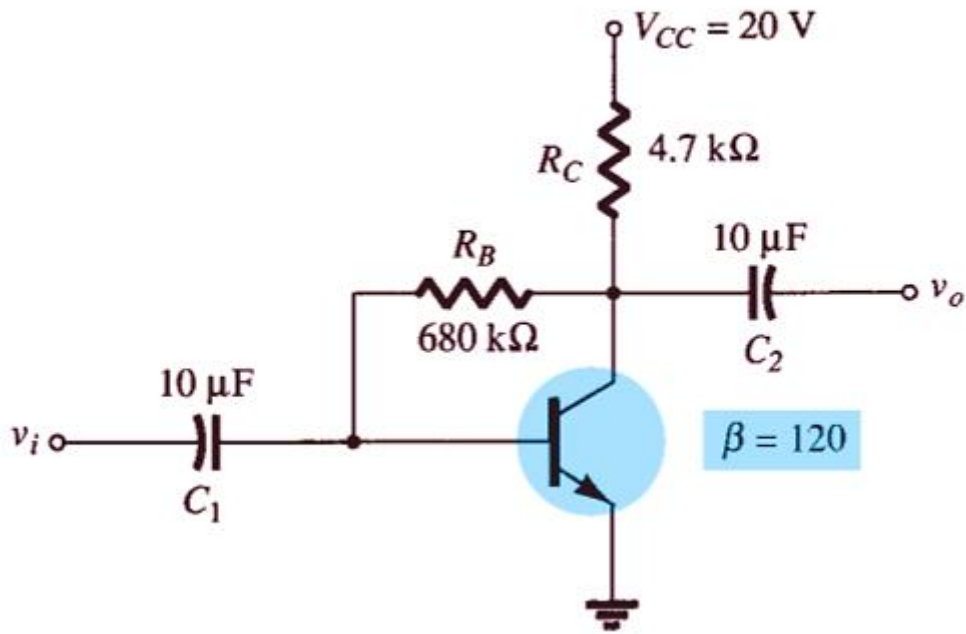
$$V_C = V_{CC} - I_C' R_C \cong V_{CC} - I_C R_C$$

$$= 18 \text{ V} - (2.66 \text{ mA})(3.3 \text{ k}\Omega)$$

$$= 18 \text{ V} - 8.78 \text{ V}$$

$$= 9.22 \text{ V}$$

Illustration (II) - >



Solution:

- a. The absence of R_E reduces the reflection of resistive levels to simply that of R_C , and the equation for I_B reduces to

$$\begin{aligned} I_B &= \frac{V_{CC} - V_{BE}}{R_B + \beta R_C} \\ &= \frac{20 \text{ V} - 0.7 \text{ V}}{680 \text{ k}\Omega + (120)(4.7 \text{ k}\Omega)} = \frac{19.3 \text{ V}}{1.244 \text{ M}\Omega} \\ &= \mathbf{15.51 \mu\text{A}} \end{aligned}$$

$$\begin{aligned} I_{CQ} &= \beta I_B = (120)(15.51 \mu\text{A}) \\ &= \mathbf{1.86 \text{ mA}} \end{aligned}$$

$$\begin{aligned} V_{CEQ} &= V_{CC} - I_C R_C \\ &= 20 \text{ V} - (1.86 \text{ mA})(4.7 \text{ k}\Omega) \\ &= \mathbf{11.26 \text{ V}} \end{aligned}$$

b.

$$V_B = V_{BE} = \mathbf{0.7 \text{ V}}$$

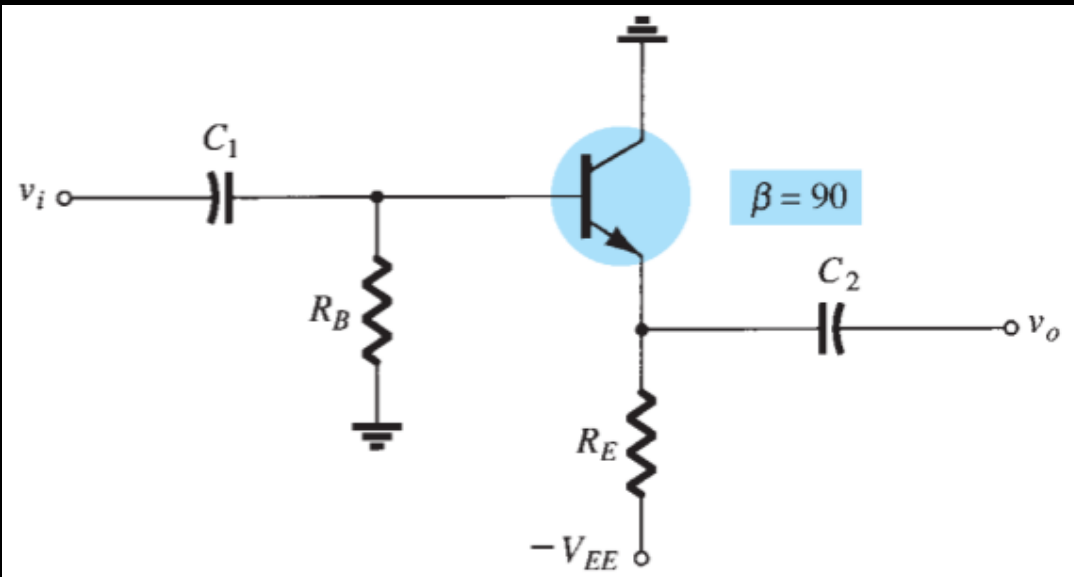
$$V_C = V_{CE} = \mathbf{11.26 \text{ V}}$$

$$V_E = \mathbf{0 \text{ V}}$$

$$\begin{aligned} V_{BC} &= V_B - V_C = 0.7 \text{ V} - 11.26 \text{ V} \\ &= \mathbf{-10.56 \text{ V}} \end{aligned}$$

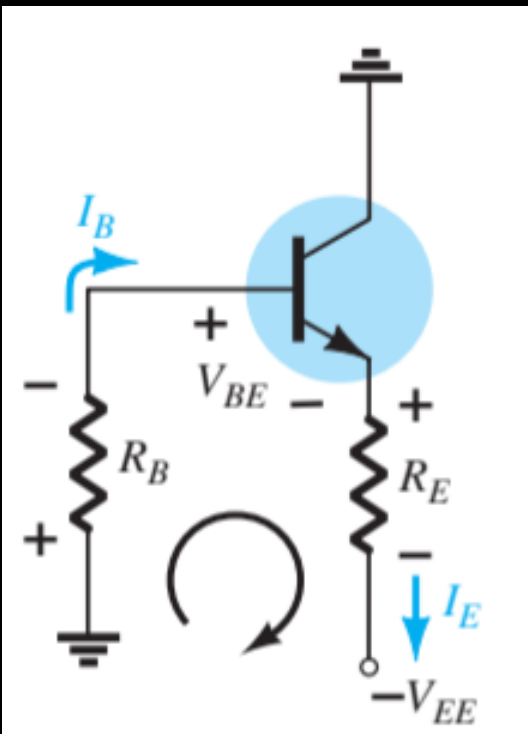
5. EMITTER FOLLOWER conf.

[DC BIASING]



Limitations :

- Voltage Gain Less Than 1 :** Unlike traditional BJT amplifiers, the emitter follower exhibits a voltage gain of less than 1 (typically close to unity). This makes it unsuitable for applications where voltage amplification is required.
- Higher Input Impedance, Lower Output Impedance :** Compared to a common-emitter configuration, the emitter follower has a high input impedance and a low output impedance. This can be advantageous for certain applications but might necessitate additional circuitry for impedance matching in other cases.



The dc equivalent of the network of Fig. 4.46 appears in Fig. 4.47

Applying Kirchhoff's voltage rule to the input circuit will result in

$$-I_B R_B - V_{BE} - I_E R_E + V_{EE} = 0$$

and using $I_E = (\beta + 1)I_B$

$$I_B R_B + (\beta + 1)I_B R_E = V_{EE} - V_{BE}$$

so that

$$I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E}$$

For the output network, an application of Kirchhoff's voltage law will result in

$$-V_{CE} - I_E R_E + V_{EE} = 0$$

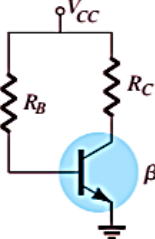
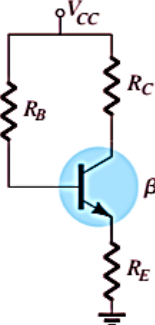
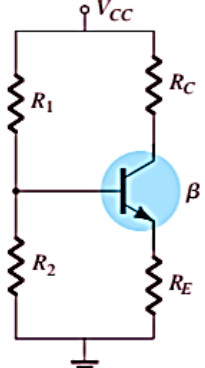
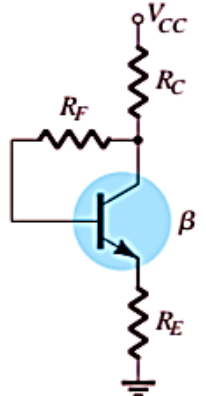
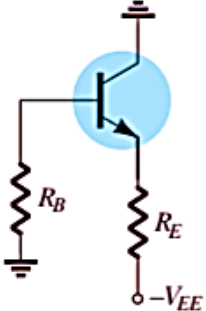
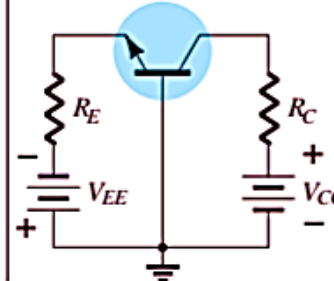
and

$$V_{CE} = V_{EE} - I_E R_E$$

❑ Advantages:

- **Voltage Buffering:** The emitter follower excels as a voltage buffer. Its high input impedance minimizes loading effects on the preceding stage, while its low output impedance provides a good driving capability for the following stage. This makes it ideal for isolating stages with high source impedance and driving low-impedance loads.
- **Improved Linearity:** The negative feedback inherent in the emitter follower configuration helps to improve the linearity of the amplified signal. This is beneficial in applications where minimizing distortion and maintaining a faithful reproduction of the input signal is essential, such as audio pre-amplifiers or buffer circuits for analog-to-digital converters (ADCs).
- **Impedance Matching:** By transforming a high-impedance source to a low-impedance output, the emitter follower acts as an impedance converter. This can be crucial for interfacing circuits with different impedance characteristics, ensuring optimal signal transfer and power delivery.

... SUMMARY ...

Type	Configuration	Pertinent Equations
Fixed-bias		$I_B = \frac{V_{CC} - V_{BE}}{R_B}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $V_{CE} = V_{CC} - I_C R_C$
Emitter-bias		$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $R_i = (\beta + 1)R_E$ $V_{CE} = V_{CC} - I_C(R_C + R_E)$
Voltage-divider bias		EXACT: $R_{Th} = R_1 R_2, E_{Th} = \frac{R_2 V_{CC}}{R_1 + R_2}$ $I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $V_{CE} = V_{CC} - I_C(R_C + R_E)$ APPROXIMATE: $\beta R_E \geq 10R_2$ $V_B = \frac{R_2 V_{CC}}{R_1 + R_2}, V_E = V_B - V_{BE}$ $I_E = \frac{V_E}{R_E}, I_B = \frac{I_E}{\beta + 1}$ $V_{CE} = V_{CC} - I_C(R_C + R_E)$
Collector-feedback		$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta(R_C + R_E)}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $V_{CE} = V_{CC} - I_C(R_C + R_E)$
Emitter-follower		$I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $V_{CE} = V_{EE} - I_E R_E$
Common-base		$I_E = \frac{V_{EE} - V_{BE}}{R_E}$ $I_B = \frac{I_E}{\beta + 1}, I_C = \beta I_B$ $V_{CE} = V_{EE} + V_{CC} - I_E(R_C + R_E)$ $V_{CB} = V_{CC} - I_C R_C$